

LIMS Print Pipeline

Operations Manual — every command, every prompt, every value, in order

How to use this manual

Work through the parts in order. **Parts 1–4** are one-time setup (cloud, hub, clients). **Parts 5–6** are day-to-day use. **Parts 7–9** are password management, the single-PC test, and maintenance. Every box in a [monospace font](#) is typed exactly as shown; every table row marked **Prompt** is what the tool asks and what you answer.

The three machines (who runs what)

Tier	Machine	Internet?	What runs here
Cloud LIMS	Supabase + a browser (anywhere)	yes	Database + web app; testers create registrations
Central hub	one desktop, e.g. 192.168.1.172	yes	hub\run.bat; hosts the limsDocs share
Clients (x60)	each lab PC	no (LAN only)	install.bat; prints to the hub

Everything you will ever run (master list)

Run this	On	Does what
hub\run.bat	hub	Start the hub server (keep the window open)
schema.sql (Supabase SQL editor)	cloud	Create the LIMS database (one time)
npm install / npm run dev	cloud/lims/web	Start the tester web app
install.bat	each client	GUI wizard: create a printer + enroll it with the hub
add-printer.bat	each client	GUI wizard: add another printer to an installed client
status.bat	client	Show printers, device names, recent log
fix-queue.bat	client	Repair a stuck queue + write diagnostics
change-password.bat	client	Change ONE printer's PDF password (super-admin)
view-password.bat	client	Reveal a printer's PDF password (super-admin)
set-super-admin.bat	client	Rotate the super-admin password
set-password.bat	client	Set the GLOBAL PDF password (all printers)
decode-pdf.bat	anywhere	Decrypt an encrypted PDF with its password
set-id.bat / print-register.bat	client	Manual/batch registration fallback
uninstall.bat	client	Remove printers, port, files

Do this once, on any machine with a browser.

1.1 Create the project & database

Step	Action / value to enter
1	supabase.com → create project. Copy Project URL, anon key, service_role key (Settings → API).
2	SQL Editor → paste lims/supabase/schema.sql → Run (idempotent, safe to re-run).
3	Authentication → enable Email provider; add a login (email + password) per tester.
4	(optional) add more device types: run insert into device_types (id) values ('xrf');

1.2 Start the tester web app

```
cd lims\web
copy .env.example .env
:: edit .env and set these two lines from Settings -> API:
VITE_SUPABASE_URL=https://YOUR-PROJECT-REF.supabase.co
VITE_SUPABASE_ANON_KEY=YOUR-ANON-KEY
npm install
npm run dev :: dev server (or: npm run build to host dist\)
```

Never put the service_role key in .env

The web app uses the ANON key only. The service_role key belongs solely on the central desktop (Part 2). Putting it in the web app would expose full database access to every browser.

1.3 Web app entries (what testers type)

Screen	Field	Value
Sign in	Email / Password	the login you created in step 3
Registrations	Registration number	e.g. R-2026-0001
Registrations	Device type	pick gcms / lcms / icpms ...
Registrations	Product / description	free text (optional)
Registrations	Tests	type a name, press Enter to add each chip (e.g. rx-test)
Documents	Filters / Download	filter by reg no / device; Download opens the PDF

On the central desktop (the one with internet + the limsDocs share, e.g. 192.168.1.172). Requires **uv** (installed automatically the first time, or from docs.astral.sh/uv).

2.1 Prepare the shared folder

```
mkdir C:\limsDocs
:: Share it on the LAN so clients see \\192.168.1.172\limsDocs
:: (right-click folder -> Properties -> Sharing -> grant READ to lab users)
```

2.2 Start the hub

```
cd hub
run.bat :: double-click it, or run in a Command Prompt
:: leave this window OPEN - closing it stops the hub
```

The window prints two secrets — write them down. They persist, so they stay the same on every restart:

Secret	Used for
ADMIN TOKEN	unlocks the dashboard + all /api endpoints
ENROLL KEY	given to each client's install.bat to register its printer

2.3 Configure the hub (dashboard)

Where	Field	Value to enter
top bar	admin token	paste the ADMIN TOKEN, click Save
Settings	limsDocs directory	C:\limsDocs (the LOCAL path of the shared folder)
Settings	Supabase URL	your Project URL (blank = local mode, no cloud)
Settings	Supabase service role key	the service_role key (stored DPAPI-encrypted here only)
Settings	Catalog poll seconds	2 (default)
Settings		click Save settings

Open the dashboard at <http://localhost:8000/> on the hub itself, or <http://192.168.1.172:8000/> from another PC. Allow inbound TCP 8000 in the firewall.

2.4 Hub tabs (what each does)

Tab	Use it to
Devices	See enrolled printers; Show enroll key; Revoke a device's token
Documents	See every job (filed / held / forwarded / forward_failed); download; auto-refresh
Held	Assign a Registration + Test to a held job → files & forwards it
Catalog	Add/remove device types + registrations (local mode); read-only when synced from Supabase
Settings	limsDocs dir, Supabase URL + key, poll interval

2.5 Hub environment overrides (optional)

Variable	Default	Effect
HUB_HOST	0.0.0.0	bind address (must be reachable from the LAN)
HUB_PORT	8000	listening port
HUB_LIMSDOCS_DIR	(from Settings)	override the limsDocs path

Variable	Default	Effect
SUPABASE_SERVICE_KEY	(from Settings)	override the stored service key
HUB_DATA_DIR	hub/data	state dir (tokens, DB, held) - for tests

On each of the 60 lab PCs. Double-click `install.bat` and approve the **UAC / Administrator** prompt. A **click-through wizard** opens (Next → Next → Install; Ctrl+V paste works in every box). Pick **LIMS hub printer**, fill the fields below, click **Test hub & load device types** to check the enroll key, then Next → Next → **Install** and watch the progress in the window. First run downloads Ghostscript, mfilemon and Python once (needs internet that first time only, or place mfilemon-setup.exe in vendor).

3.1 Every wizard field, in order

Field	What to type	Notes
Mode	LIMS hub printer	the Legacy option is for direct-URL printers only
Hub base URL	<code>http://192.168.1.172:8000</code>	BASE ONLY — never an <code>/ingest/...</code> link (the wizard rejects those)
Enroll key	the hub's ENROLL KEY	paste it; "Test hub" confirms it and loads device types
Printer name	e.g. GCMS Printer	how it appears in the Windows print dialog
Device name	e.g. gcms-01	must be UNIQUE across all clients (409 if taken)
Device type	gcms / lcms / icpms	dropdown, loaded from the hub
PDF password (+ repeat)	a password, or blank	blank = no encryption for this printer
Catalog share folder	leave the auto-filled value	<code>\\limsDocs\vcpl\catalog</code>

The #1 mistake

Hub base URL is `http://HOST:8000` with NOTHING after the port. An `/ingest/<token>` link is a per-printer address the hub gives back automatically — it is not the hub URL and enrollment will 404 if you paste it.

3.2 Scripted install (mass rollout, no prompts)

```
powershell -ExecutionPolicy Bypass -File setup.ps1 -Action install ^
-PrinterName "GCMS Printer" -HubUrl http://192.168.1.172:8000 ^
-DeviceName gcms-01 -DeviceType gcms -EnrollKey ^
-Password -CatalogShareDir \\192.168.1.172\limsDocs\vcpl\catalog
```

Legacy direct-URL printer (no hub): pass `-Url https://your-endpoint` instead of the hub options — it behaves like the original tool.

3.3 Confirm it worked

Check	Expected
Installer output	"Printer '<name>' is ready"
Hub → Devices tab	the new device_name is listed
status.bat (on the client)	printer shown with [device: name / type]

4.1 Create a registration (tester)

In the LIMS web app (or, in local mode, the hub's Catalog tab): add a registration number, its device type, and one or more tests. Within ~2 seconds it reaches the hub and every client's dropdowns.

4.2 Print (lab user)

Step	Action	Result
1	Any app → Print → choose the printer	the registration dialog opens
2	Pick the Registration (searchable)	the Test dropdown fills
3	Pick the Test → Attach & print	PDF is sent to the hub
4	(if you click Skip)	job is HELD at the hub until assigned

4.3 Where the file lands

```
\\192.168.1.172\limsDocs\____.pdf
:: and, in Supabase mode, in the LIMS Documents page within seconds
```

4.4 Handle held jobs (operator)

Step	Action
1	Hub dashboard → Held tab
2	In the job's row pick a Registration, then a Test
3	Click Assign → the file moves into the tree and forwards to the cloud

Factory super-admin password: Citta@26252423 — rotate it on day one

Run set-super-admin.bat on each client and set your own. It is stored only as a PBKDF2-SHA256 hash; the factory value works until you change it. It gates change-password / view-password / set-super-admin.

5.1 Password commands (all run on the client, elevated)

Command	Prompts (in order)	Effect
change-password.bat	printer name; super-admin password; new password (twice)	set/replace ONE printer's PDF password (DPAPI at rest); blank new = disable
view-password.bat	printer name; super-admin password	shows that printer's password on screen only
set-super-admin.bat	current super-admin; new (twice)	rotate the super-admin password
set-password.bat	passphrase (blank = off)	GLOBAL password for printers without their own
decode-pdf.bat	pick file(s); password	decrypt an encrypted PDF to -decrypted.pdf

5.2 Security guarantees (no action needed — built in)

Protection	How
Per-device tokens	each printer's ingest token is unique; revoke in Devices tab
Passwords at rest	DPAPI machine-scope on clients; PBKDF2 hash for super-admin
Cloud key on-site	service_role key stays DPAPI-encrypted on the hub only
No path escape	every folder segment from a client is sanitized
Fail-closed	encryption on + no password/qpdf → job held in failed\, never sent clear
Locked install	%ProgramData%\VirtualCloudPrinter is SYSTEM+Admins only

Test it on ONE PC (before the real rollout)

Proves the whole pipeline on a single Windows PC in local mode — no LAN, no Supabase. ~15 min.

#	Run / do	Expected
1	cd hub then run.bat	window shows ADMIN TOKEN + ENROLL KEY; dashboard at http://localhost:8000/
2	Browser → localhost:8000 → paste admin token → Settings → lmsDocs = C:\lmsDocs → Save	local mode, no error
3	Catalog tab → add R-TEST-1, gcms, tests rx-test, ry-test	R-TEST-1 in the table
4	install.bat → wizard: hub http://127.0.0.1:8000/ + enroll key → Test hub → Next → printer lmsPrinter, device gcms-01, type gcms, blank pw → Next → Install	wizard shows DONE; device in Devices tab
5	Notepad → Print → lmsPrinter	registration dialog shows R-TEST-1
6	Pick R-TEST-1 → rx-test → Attach & print	dialog closes
7	Look in C:\lmsDocs\gcms-01\gcms\R-TEST-1\rx-test\ + Documents tab	a .pdf there; status filed
8	Print again → click Skip	Held tab shows it; Assign a reg+test → it files

If the dialog never shows or nothing files

Run fix-queue.bat (elevated). Read C:\ProgramData\VirtualCloudPrinter\log.txt for the per-job trace and the hub window for what it received. Unposted PDFs are always kept in the client's failed\ folder.

7.1 Routine checks

Run	Shows / does
status.bat (client)	port monitor, printers, device names, last 12 log lines
Hub → Documents	live job statuses; filter by filed/held/forwarded/forward_failed
Hub → Devices	enrolled clients + last-seen; revoke a token
C:\ProgramData\VirtualCloudPrinter\log.txt	per-job client trace
client_failed\ folder	PDFs that could not be uploaded (never lost)

7.2 Problem → fix

Symptom	Cause & fix
Dialog dropdown empty	no open registration of that device type — add one (LIMS / Catalog tab)
"Catalog unavailable"	client can't reach hub & share fallback unreadable — check hub up, port 8000, .vcp\catalog files
Enroll fails 409	device name already used — pick a unique one
Enroll fails 401	wrong enroll key — Devices tab → Show enroll key
Wrong app on :8000	title says "Receiver Simulator" — that's the old tool; run hub\run.bat instead
Job stuck 'held'	expected without a valid registration — assign it in the Held tab
'forward_failed'	hub can't reach Supabase — check internet + keys; it retries automatically
Nothing prints	fix-queue.bat; read log.txt; check failed\

7.3 Remove a client install

```
uninstall.bat :: removes printers, the port, and %ProgramData%\VirtualCloudPrinter
```

Reference documents in the project

ARCHITECTURE.md (interface contracts) · LAN-SETUP.md (deployment runbook) · LIMS-Pipeline-Reference.pdf (architecture + flows + security) · this manual (every command).